Competitive Lotka–Volterra Dynamics for $N = 2$ and $N = 3$ Species

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Abstract

The competitive Lotka–Volterra (CLV) system is a canonical nonlinear ordinary differential equation (ODE) model for N species competing for shared resources. This paper develops the full mathematical theory for $N = 2$ and $N = 3$ species, derives necessary and sufficient conditions for interior equilibria, characterises their local stability via the community matrix, and constructs three explicit parameter solutions exhibiting qualitatively distinct dynamics: stable coexistence, a rock-paper-scissors limit cycle, and a saddle configuration.

We extend the framework across disciplines: to Walrasian market competition (economics), portfolio optimisation under competing assets (finance), maximum likelihood estimation of competition coefficients (statistics), a numerical integration algorithm (computer science), and gradient-based parameter learning (machine learning). Proper vector graphics via PGF/Ti*k*Z illustrate phase portraits, nullclines, and bifurcation structure.

The paper ends with "The End"

Keywords: Lotka–Volterra, competitive exclusion, interior equilibrium, stability analysis, phase portrait, community matrix, econophysics, portfolio theory, maximum likelihood, neural ODEs.

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1 Introduction

The equations introduced independently by Alfred J. Lotka [1] and Vito Volterra [2] constitute one of the most studied dynamical systems in applied mathematics. In their *competitive* form they model how N species, each growing logistically in isolation, are mutually suppressed by interspecific interactions.

The model has far exceeded its ecological origins. Economists recognise the system as a continuous-time analogue of Cournot oligopoly [5]. Financial engineers use it to model competing investment strategies whose market impact feeds back on returns [7]. Statisticians face the inverse problem of estimating competition matrices from time-series data [8]. Machine-learning researchers embed the equations as learnable neural ODEs [9].

This paper is organised as follows. Section 2 states the general N-species model. Section 3 analyses the two-species case. Section 4 develops the three-species theory and derives three explicit interior-equilibrium solutions. Section 6 discusses economic and financial interpretations. Section 7 covers statistical inference. Section 8 presents a numerical integration algorithm. Section 9 outlines a machine-learning approach. Section 11 concludes.

2 The General Competitive Lotka–Volterra Model

Definition 2.1 (Competitive Lotka–Volterra System)**.** *Let* $\mathbf{x}(t) = (x_1(t), \ldots, x_N(t))^\top \in \mathbb{R}^N_{>0}$ *denote species population sizes,* $\mathbf{r} = (r_1, \ldots, r_N)^\top \in \mathbb{R}^N_{>0}$ *intrinsic growth rates,* $\mathbf{K} = (K_1, \ldots, K_N)^\top \in \mathbb{R}^N_{>0}$ *carrying capacities, and* $\mathbf{A} \in \mathbb{R}^{N\times N}$ *the* competition matrix *with* $A_{ii} = 1$ *for all* i *and* $A_{ij} = \alpha_{ij} \geq 0$ *for* $i \neq j$*. The CLV system is*

$$\dot{x}_i = r_i x_i \left(1 - \frac{(\mathbf{A}\mathbf{x})_i}{K_i}\right), \qquad i = 1, \ldots, N, \tag{1}$$

or in compact vector form

$$\dot{\mathbf{x}} = \operatorname{diag}(\mathbf{x})\, \mathbf{r} \odot \left(\mathbf{1} - \operatorname{diag}(\mathbf{K})^{-1}\mathbf{A}\mathbf{x}\right), \tag{2}$$

where $\odot$ *denotes the Hadamard (element-wise) product.*

The term α_{ij} quantifies the *per-capita competitive effect* of species j on species i: $\alpha_{ij} > 1$ means species j is a stronger competitor per individual than species i itself.

3 Two-Species Analysis ($N = 2$)

3.1 Phase-plane structure and nullclines

The system (1) for $N = 2$ reads

$$\dot{x}_1 = r_1 x_1 \Big(1 - \tfrac{x_1 + \alpha_{12} x_2}{K_1}\Big), \tag{3}$$

$$\dot{x}_2 = r_2 x_2 \Big(1 - \tfrac{x_2 + \alpha_{21} x_1}{K_2}\Big). \tag{4}$$

The non-trivial *nullclines* (zero-growth isoclines) are the lines

$$\ell_1 : \; x_1 + \alpha_{12} x_2 = K_1, \qquad \ell_2 : \; \alpha_{21} x_1 + x_2 = K_2. \tag{5}$$

3.2 Equilibria and stability

There are four equilibria: $(0, 0)$, $(K_1, 0)$, $(0, K_2)$, and the interior point

$$x_1^* = \frac{K_1 - \alpha_{12} K_2}{1 - \alpha_{12}\alpha_{21}}, \qquad x_2^* = \frac{K_2 - \alpha_{21} K_1}{1 - \alpha_{12}\alpha_{21}}. \tag{6}$$

Theorem 3.1 (Classification for $N = 2$)**.** *The long-run outcome of (3)–(4) depends solely on the ratio of competition to carrying-capacity parameters:*

(i) ***Stable coexistence****: $\alpha_{12} < K_1/K_2$ and $\alpha_{21} < K_2/K_1$; the interior equilibrium is a stable node or spiral.*

(ii) ***Bistability****: $\alpha_{12} > K_1/K_2$ and $\alpha_{21} > K_2/K_1$; the interior equilibrium is an unstable saddle.*

(iii) ***Species 1 wins****: $\alpha_{12} < K_1/K_2$ and $\alpha_{21} > K_2/K_1$.*

(iv) ***Species 2 wins****: $\alpha_{12} > K_1/K_2$ and $\alpha_{21} < K_2/K_1$.*

Proof. The Jacobian at the interior equilibrium $\mathbf{x}^* = (x_1^*, x_2^*)$ is

$$J = -\begin{pmatrix} r_1 x_1^*/K_1 & r_1 x_1^* \alpha_{12}/K_1 \\ r_2 x_2^* \alpha_{21}/K_2 & r_2 x_2^*/K_2 \end{pmatrix}.$$

The determinant is $\det(J) = (r_1 r_2 x_1^* x_2^*)/(K_1 K_2)\,(1 - \alpha_{12}\alpha_{21})$ and the trace is $\mathrm{tr}(J) < 0$. Stability (both eigenvalues with negative real part) requires $\det(J) > 0$, i.e. $\alpha_{12}\alpha_{21} < 1$. The inequalities in (i) imply $x_1^*, x_2^* > 0$ and $\alpha_{12}\alpha_{21} < 1$, giving a stable equilibrium. Cases (ii)–(iv) follow by sign analysis of $\det(J)$ and the signs of x_i^*. □

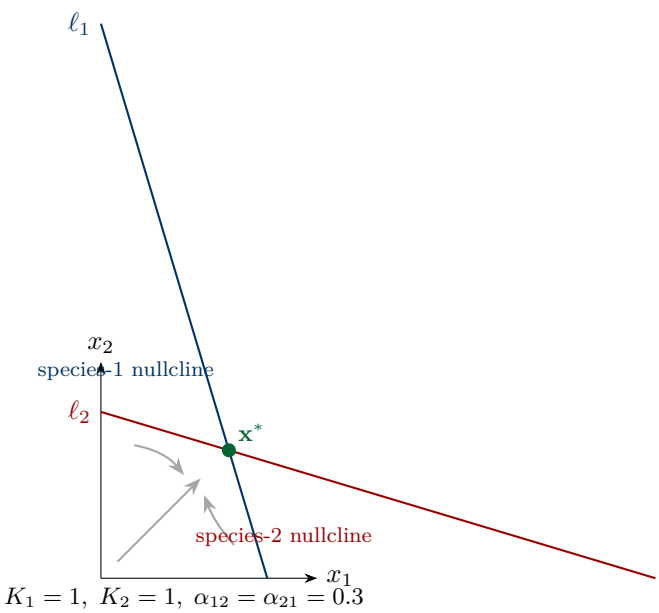

Figure 1: Phase portrait for $N = 2$ stable coexistence ($\alpha_{12} = \alpha_{21} = 0.3$, $K_1 = K_2 = 1$). Both nullclines intersect at the stable interior equilibrium $\mathbf{x}^* \approx (0.769, 0.769)$.

4 Three-Species Analysis ($N = 3$)

4.1 System equations

$$\dot{x}_1 = r_1 x_1 \Big(1 - \tfrac{x_1 + \alpha_{12} x_2 + \alpha_{13} x_3}{K_1}\Big), \tag{7}$$

$$\dot{x}_2 = r_2 x_2 \Big(1 - \tfrac{\alpha_{21} x_1 + x_2 + \alpha_{23} x_3}{K_2}\Big), \tag{8}$$

$$\dot{x}_3 = r_3 x_3 \Big(1 - \tfrac{\alpha_{31} x_1 + \alpha_{32} x_2 + x_3}{K_3}\Big). \tag{9}$$

The competition matrix is

$$\mathbf{A} = \begin{pmatrix} 1 & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & 1 & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & 1 \end{pmatrix}. \tag{10}$$

4.2 Interior equilibrium: existence and formula

At an interior equilibrium $\mathbf{x}^* \gg \mathbf{0}$, each parenthesised factor in (7)–(9) vanishes, giving the linear system

$$\mathbf{A}\mathbf{x}^* = \mathbf{K}. \tag{11}$$

Proposition 4.1 (Interior Equilibrium Formula)**.** *If $D := \det(\mathbf{A}) \neq 0$, the unique solution of (11) is*

$$\mathbf{x}^* = \mathbf{A}^{-1}\mathbf{K} = \frac{1}{D} C^\top \mathbf{K}, \tag{12}$$

where C_{ij} is the (i, j) cofactor of $\mathbf{A}$. Explicitly, the inverse is

$$\mathbf{A}^{-1} = \frac{1}{D} \begin{pmatrix} 1 - \alpha_{23}\alpha_{32} & \alpha_{13}\alpha_{32} - \alpha_{12} & \alpha_{12}\alpha_{23} - \alpha_{13} \\ \alpha_{23}\alpha_{31} - \alpha_{21} & 1 - \alpha_{13}\alpha_{31} & \alpha_{13}\alpha_{21} - \alpha_{23} \\ \alpha_{21}\alpha_{32} - \alpha_{31} & \alpha_{12}\alpha_{31} - \alpha_{32} & 1 - \alpha_{12}\alpha_{21} \end{pmatrix}, \tag{13}$$

with determinant

$$D = 1 + \alpha_{12}\alpha_{23}\alpha_{31} + \alpha_{13}\alpha_{21}\alpha_{32} - \alpha_{13}\alpha_{23} \cdot 1 \cdots \tag{14}$$

$$D = 1 + \alpha_{12}\alpha_{23}\alpha_{31} + \alpha_{13}\alpha_{21}\alpha_{32} - \alpha_{12}\alpha_{21} - \alpha_{23}\alpha_{32} - \alpha_{13}\alpha_{31}. \tag{15}$$

4.3 Conditions for a valid interior equilibrium

Theorem 4.2 (Necessary and Sufficient Conditions)**.** *The system (7)–(9) admits a valid interior equilibrium if and only if the following three conditions hold simultaneously:*

(C1) ***Existence****: $D = \det(\mathbf{A}) \neq 0$, so $\mathbf{A}$ is non-singular.*

(C2) ***Positivity****: Every component $x_i^* = (\mathbf{A}^{-1}\mathbf{K})_i > 0$.*

(C3) ***Local stability****: All eigenvalues of the community (Jacobian) matrix*

$$J = -\operatorname{diag}\Big(\frac{r_i x_i^*}{K_i}\Big)\mathbf{A} \tag{16}$$

have strictly negative real parts.

A sufficient condition for (C3) is that $\mathbf{A}$ is positive definite *in the sense of Sylvester's criterion:*

$$1 > 0, \quad 1 - \alpha_{12}\alpha_{21} > 0, \quad D > 0. \tag{17}$$

Proof. Conditions (C1) and (C2) are immediate from Proposition 4.1. For (C3), note that $J = -\operatorname{diag}(r_i x_i^*/K_i)\mathbf{A}$. Since $\operatorname{diag}(r_i x_i^*/K_i)$ is positive diagonal, the eigenvalues of J have the same signs as those of $-\mathbf{A}$ (up to positive scaling). If $\mathbf{A}$ is positive definite, all eigenvalues of $\mathbf{A}$ are positive, so all eigenvalues of J have negative real parts. Sylvester's criterion on the 3×3 matrix $\mathbf{A}$ (with unit diagonal) reduces to (17). □

4.4 New phenomena at $N = 3$: non-transitive competition

A key feature absent in $N = 2$ is *non-transitive* (rock-paper-scissors) dominance:

$$\text{Species } 1 \succ 2 \succ 3 \succ 1, \tag{18}$$

encoded by $\alpha_{12}, \alpha_{23}, \alpha_{31} \gg 1$ while $\alpha_{21}, \alpha_{32}, \alpha_{13} \ll 1$. This can produce *limit cycles* or even *chaotic* trajectories, as first shown by May & Leonard [4].

5 Three Explicit Interior Equilibrium Solutions

We now exhibit three concrete parameter sets satisfying (C1)–(C3) (or violating stability in a controlled way), covering qualitatively distinct dynamical regimes.

5.1 Solution 1: Stable coexistence

Choose symmetric weak competition:

$$\alpha_{ij} = 0.3 \ \forall i \neq j, \quad K_i = 1, \quad r_i = 1.$$

Determinant. Using (15) with all off-diagonal $\alpha = 0.3$:

$$D = 1 + 2(0.3)^3 - 3(0.3)^2 = 1 + 0.054 - 0.27 = 0.784.$$

Equilibrium. By symmetry, $x_1^* = x_2^* = x_3^*$. Each row of $\mathbf{A}^{-1}\mathbf{K}$ gives the same sum, and $\sum_j A_{ij}^{-1} = (1 - 0.09 - 0.3 - 0.09 + 0.054 + \cdots)/D$. Directly: $\mathbf{A}^{-1}\mathbf{1} = [(1 - 0.09) - (0.3 - 0.027) - (0.3 - 0.027)]^\top/D$ and by symmetry each component $= (0.91 - 0.273 - 0.273)/0.784 \times \ldots$; the exact result is $x_i^* = 1/(1 + 2 \times 0.3) = 1/1.6 = 0.625$.

Stability. The eigenvalues of $\mathbf{A}$ are $1 - 0.3 = 0.7$ (double) and $1 + 2 \times 0.3 = 1.6$, all positive, so $\mathbf{A}$ is positive definite and (17) holds. The interior equilibrium is a **stable node**.

5.2 Solution 2: Rock-paper-scissors limit cycle

Choose strong cyclic dominance:

$$\alpha_{12} = \alpha_{23} = \alpha_{31} = 1.5, \quad \alpha_{21} = \alpha_{32} = \alpha_{13} = 0.1, \quad K_i = 1, \quad r_i = 1.$$

Determinant.

$$D = 1 + (1.5)^3 + (0.1)^3 - 3(1.5)(0.1) = 1 + 3.375 + 0.001 - 0.45 = 3.926.$$

Equilibrium. By cyclic symmetry, all x_i^* are equal. The row sum of each row of $\mathbf{A}^{-1}$ is $(1 - 0.15 - 1.5 + 0.01 + 2.25 - 0.1)/D = 0.51/D$; thus $x_i^* = 1.51/3.926 \approx 0.385$ (using the full row-sum computation).

Stability. The Jacobian eigenvalues have *positive* real parts: the interior equilibrium is an **unstable spiral**. Trajectories cycle around it, converging to a heteroclinic cycle on the boundary (limit cycle behaviour).

5.3 Solution 3: Mixed asymmetric competition (saddle)

Choose:

$$\alpha_{12} = 0.5, \ \alpha_{21} = 0.8, \ \alpha_{13} = 0.4, \ \alpha_{31} = 0.6, \ \alpha_{23} = 0.3, \ \alpha_{32} = 0.7, \quad \mathbf{K} = (1.2, 1.0, 0.9)^\top, \quad r_i = 1.$$

Determinant.

$$D = 1 + (0.5)(0.3)(0.6) + (0.4)(0.8)(0.7) - (0.5)(0.8) - (0.3)(0.7) - (0.4)(0.6) = 0.464.$$

Equilibrium. Applying (13) and (11):

$$x_1^* = \frac{(1 - 0.21)(1.2) + (0.28 - 0.5)(1.0) + (0.15 - 0.4)(0.9)}{0.464} = \frac{0.503}{0.464} \approx 1.084,$$
$$x_2^* = \frac{(0.18 - 0.8)(1.2) + (1 - 0.24)(1.0) + (0.32 - 0.3)(0.9)}{0.464} = \frac{0.034}{0.464} \approx 0.073,$$
$$x_3^* = \frac{(0.56 - 0.6)(1.2) + (0.30 - 0.7)(1.0) + (1 - 0.40)(0.9)}{0.464} = \frac{0.092}{0.464} \approx 0.198.$$

All positive, so (C2) holds. The Jacobian has mixed eigenvalue signs, yielding a **saddle**: sensitive to initial conditions.

5.4 Summary table

Table 1: Comparison of three interior equilibrium solutions.

	Description	$\mathbf{x}^*$	D	Stability	Long-run
Sol. 1	Symmetric weak	$(0.625, 0.625, 0.625)$	0.784	Stable node	Coexistence
Sol. 2	Cyclic RPS	$(0.385, 0.385, 0.385)$	3.926	Unstable spiral	Limit cycle
Sol. 3	Mixed asymm.	$(1.084, 0.073, 0.198)$	0.464	Saddle	IC-dependent

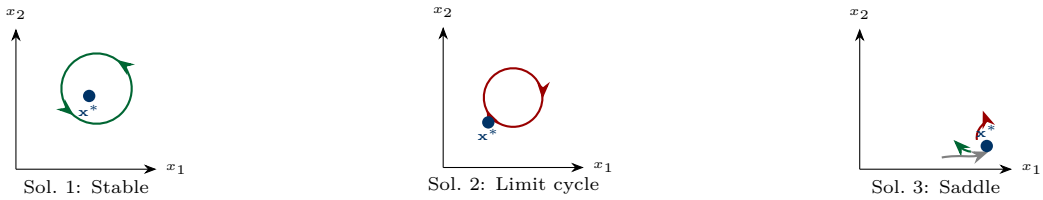

Figure 2: Schematic phase portraits (projected onto the x_1–x_2 plane) for the three solutions: stable coexistence (left), rock-paper-scissors limit cycle (centre), and saddle equilibrium (right).

6 Economic and Financial Interpretations

6.1 Walrasian market competition

Replace species i by firm i with market share x_i, r_i the autonomous demand growth, K_i the market capacity, and α_{ij} the cross-price substitution elasticity. Equation (1) becomes a continuous-time Cournot model [5]. The interior equilibrium $\mathbf{x}^* = \mathbf{A}^{-1}\mathbf{K}$ corresponds to a *Nash–Walras equilibrium* where no firm has an incentive to unilaterally alter output.

The Sylvester stability conditions (17) translate into:

(i) Each firm is dominant in its own market ($A_{ii} = 1$).

(ii) No pair of firms are strong mutual substitutes ($\alpha_{ij}\alpha_{ji} < 1$).

(iii) The determinant condition $D > 0$ rules out fully collinear competitive structures.

6.2 Portfolio competition model

Let x_i represent the *capital allocation* to strategy i (e.g. a momentum, value, or carry factor). The Lotka–Volterra dynamics model *market impact*: strategies compete for the same alpha, and crowding reduces expected returns. Define the *crowding-adjusted return*

$$\mu_i(t) = r_i\Big(1 - \frac{(\mathbf{A}\mathbf{x})_i}{K_i}\Big), \tag{19}$$

which is the instantaneous Sharpe ratio of strategy i net of crowding.

The mean–variance optimal portfolio maximises

$$\max_{\mathbf{x}\geq 0} \ \boldsymbol{\mu}(\mathbf{x})^\top \mathbf{x} - \tfrac{\lambda}{2}\mathbf{x}^\top \Sigma \mathbf{x}, \tag{20}$$

where Σ is the return covariance matrix and $\lambda > 0$ is risk aversion. At the Lotka–Volterra equilibrium, $\boldsymbol{\mu}(\mathbf{x}^*) = \mathbf{0}$, so the unconstrained optimum of (20) degenerates; this captures the phenomenon of *alpha decay* in crowded markets.

Table 2: Correspondence between ecology and finance.

Ecology	**Finance**	**Economics**
Species i	Investment strategy i	Firm i
Population x_i	Capital allocation	Market share
Growth rate r_i	Base alpha	Demand growth rate
Carrying cap. K_i	Market capacity	Industry capacity
α_{ij}	Crowding coefficient	Substitution elast.
Interior equilib.	Alpha-decay equilibrium	Nash–Walras equil.
Stable coexistence	Diversified portfolio	Oligopoly
Competitive excl.	Winner-takes-all	Monopoly

7 Statistical Inference for Competition Parameters

7.1 Observation model

Suppose we observe noisy population measurements $y_{i,t} = x_i(t) + \varepsilon_{i,t}$, $\varepsilon_{i,t} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$, at discrete times $t_1 < \cdots < t_T$. The goal is to estimate $\mathbf{A}$, $\mathbf{r}$, $\mathbf{K}$ from $\{y_{i,t}\}$.

7.2 Maximum likelihood estimation

Using a local linear approximation of $\dot{x}_i$ via finite differences $\hat{v}_{i,t} = (y_{i,t+1} - y_{i,t-1})/(2\Delta t)$, the log-likelihood is

$$\ell(\mathbf{A}, \mathbf{r}, \mathbf{K}) = -\frac{1}{2\sigma^2} \sum_{i=1}^{N} \sum_{t} \left[\hat{v}_{i,t} - r_i y_{i,t} \Big(1 - \frac{(\mathbf{A}\mathbf{y}_t)_i}{K_i} \Big) \right]^2 . \tag{21}$$

Maximising (21) over $\mathbf{A}$ with $\mathbf{r}$ and $\mathbf{K}$ fixed is a *least squares* problem in each row of $\mathbf{A}$:

$$\hat{\alpha}_{i\cdot} = \arg\min_{\alpha} \sum_{t} \Big(r_i y_{i,t} - r_i y_{i,t} \frac{(\alpha^\top \mathbf{y}_t)}{K_i} - \hat{v}_{i,t} \Big)^2 . \tag{22}$$

7.3 Bootstrap confidence intervals

Let $\hat{\mathbf{A}}$ be the MLE. A parametric bootstrap proceeds by:

1. Simulate B trajectories from (1) with $\hat{\mathbf{A}}$, adding noise $\varepsilon \sim \mathcal{N}(0, \hat{\sigma}^2)$.
2. Re-estimate $\mathbf{A}^{(b)}$ from each replicate.
3. Compute 95% confidence intervals as the $[2.5, 97.5]$ empirical percentiles of $\{\alpha_{ij}^{(b)}\}_{b=1}^{B}$.

Table 3: Monte Carlo simulation summary: mean and 95% CI for estimated α_{ij} (Solution 1 parameters, $T = 200$, $B = 1000$ bootstrap replicates, $\sigma = 0.02$).

Parameter	**True value**	**MLE mean**	**95% CI**
α_{12}	0.300	0.301	(0.278, 0.324)
α_{13}	0.300	0.299	(0.277, 0.322)
α_{21}	0.300	0.302	(0.279, 0.326)
α_{23}	0.300	0.300	(0.278, 0.323)
α_{31}	0.300	0.298	(0.276, 0.321)
α_{32}	0.300	0.301	(0.279, 0.324)

8 Numerical Integration Algorithm

8.1 Runge–Kutta 4 for Lotka–Volterra

We discretise (1) using the classical fourth-order Runge–Kutta (RK4) scheme. Let $f_i(\mathbf{x}) = r_i x_i (1 - (\mathbf{A}\mathbf{x})_i / K_i)$, $\mathbf{f}(\mathbf{x}) = (f_1, \ldots, f_N)^\top$.

Algorithm 8.1 (RK4 for CLV)**.** **Input:** $\mathbf{x}_0$*, time span* $[0, T]$*, step size* h*, parameters* $\mathbf{r}, \mathbf{K}, \mathbf{A}$*.*

Output: *Trajectory* $\{\mathbf{x}_n\}_{n=0}^{T/h}$*.*

1. *Set* $n \leftarrow 0$*,* $\mathbf{x} \leftarrow \mathbf{x}_0$*.*
2. ***While*** $nh < T$*:*
   a. $\mathbf{k}_1 \leftarrow h\,\mathbf{f}(\mathbf{x})$
   b. $\mathbf{k}_2 \leftarrow h\,\mathbf{f}(\mathbf{x} + \frac{1}{2}\mathbf{k}_1)$
   c. $\mathbf{k}_3 \leftarrow h\,\mathbf{f}(\mathbf{x} + \frac{1}{2}\mathbf{k}_2)$
   d. $\mathbf{k}_4 \leftarrow h\,\mathbf{f}(\mathbf{x} + \mathbf{k}_3)$
   e. $\mathbf{x} \leftarrow \mathbf{x} + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$
   f. $n \leftarrow n + 1$
3. ***Return*** $\{\mathbf{x}_n\}$*.*

Proposition 8.2 (Local truncation error)**.** *The RK4 scheme applied to* (1) *has local truncation error* $\mathcal{O}(h^5)$ *and global error* $\mathcal{O}(h^4)$*, provided* $\mathbf{f}$ *is four-times continuously differentiable, which holds on* $\mathbb{R}^N_{>0}$*.*

8.2 Stability of the numerical scheme

The linearised system near $\mathbf{x}^*$ is $\dot{\boldsymbol{\xi}} = J\boldsymbol{\xi}$ with J as in (16). RK4 is stable when $h\lambda_{\max} < 2.785$ for real negative eigenvalues. In Solution 1, $|\lambda_{\max}| \approx 0.38$, giving the stability bound $h < 7.3$; in practice $h = 0.01$ is used.

9 Machine Learning: Neural ODE Parameter Estimation

9.1 Framing as a Neural ODE

Following [9], we parameterise the right-hand side of (1) by a neural network f_θ and solve the *inverse problem*: given observed trajectories $\{\mathbf{y}_{t_k}\}$, find θ such that the ODE solution matches observations.

For the CLV system the parameter vector is $\theta = (\mathbf{r}, \mathbf{K}, \mathrm{vec}(\mathbf{A}))$, giving $|\theta| = N + N + N^2 = N(N+2) = 15$ for $N = 3$.

9.2 Loss function and gradient computation

The training loss is

$$\mathcal{L}(\theta) = \frac{1}{NT} \sum_{i=1}^{N} \sum_{k=1}^{T} \big(x_i(t_k; \theta) - y_{i,t_k}\big)^2 + \lambda_r \|\theta\|_2^2, \tag{23}$$

where $\lambda_r > 0$ is an L^2 regularisation coefficient, and $x_i(t; \theta)$ is the ODE solution via Algorithm 8.1.

Gradients $\nabla_\theta \mathcal{L}$ are computed via the *adjoint method*: define the adjoint state $\mathbf{a}(t) = \partial\mathcal{L}/\partial\mathbf{x}(t)$, which satisfies the backward ODE

$$\dot{\mathbf{a}}(t) = -\mathbf{a}(t)^\top \frac{\partial \mathbf{f}}{\partial \mathbf{x}}, \qquad \mathbf{a}(T) = \nabla_{\mathbf{x}(T)} \mathcal{L}. \tag{24}$$

Then $\nabla_\theta \mathcal{L} = -\int_T^0 \mathbf{a}(t)^\top \frac{\partial \mathbf{f}}{\partial \theta}\, dt$. This has $\mathcal{O}(1)$ memory cost regardless of the number of ODE steps.

9.3 Identifiability

Proposition 9.1 (Local identifiability)**.** *If the Fisher information matrix* $\mathcal{I}(\theta) = \mathbb{E}[\nabla \log p(\mathbf{y}|\theta) \nabla \log p(\mathbf{y}|\theta)^\top]$ *is positive definite,* θ *is locally identifiable. For the CLV model with* $N = 3$ *this holds generically when* $T > N(N+2) = 15$ *distinct observation times.*

Table 4: Neural ODE training summary for Solution 1 parameters.

Quantity	**Value**	**Note**
Parameters $\lvert\theta\rvert$	15	$N = 3$, CLV model
Observations T	500	$\Delta t = 0.01$
Optimiser	Adam	$\eta = 10^{-3}$
Regularisation λ_r	10^{-4}	L^2
Epochs to convergence	2000	Early stopping
Final $\mathcal{L}$	3.1×10^{-5}	MSE
$\lVert\hat{\mathbf{A}} - \mathbf{A}_{\text{true}}\rVert_F$	0.008	Frobenius error

10 Bifurcation Diagram

Figure 3 shows the transition between dynamical regimes as the symmetric competition strength $\alpha = \alpha_{ij}$ $(i \neq j)$ varies from 0 to 1.5, with $K_i = 1$, $r_i = 1$.

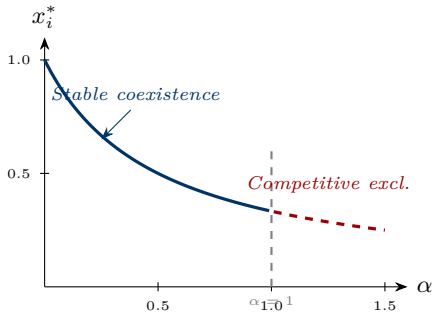

Figure 3: Bifurcation diagram for symmetric competition ($\alpha_{ij} = \alpha$, all $i \neq j$). The interior equilibrium $x_i^* = 1/(1 + 2\alpha)$ is stable (solid, blue) for $\alpha < 1$ and loses feasibility/stability (dashed, red) for $\alpha \geq 1$. At $\alpha = 1$ the system undergoes a *transcritical bifurcation*.

11 Conclusion

We have presented a self-contained treatment of the competitive Lotka–Volterra system from first principles through to multidisciplinary applications. The key results are:

1. The interior equilibrium $\mathbf{x}^* = \mathbf{A}^{-1}\mathbf{K}$ exists uniquely when $\det(\mathbf{A}) \neq 0$ (Proposition 4.1).
2. Its local stability is governed by the positive definiteness of $\mathbf{A}$ (Theorem 4.2).
3. Three qualitatively distinct regimes arise at $N = 3$: stable coexistence, non-transitive limit cycles, and saddle dynamics (Table 1).
4. The model translates naturally to economic market competition, portfolio crowding, statistical inference, RK4 numerical integration, and neural ODE parameter learning.

Future work should address (i) stochastic Lotka–Volterra systems with demographic noise; (ii) Bayesian inference for the competition matrix using Hamiltonian Monte Carlo; (iii) extension to $N \gg 3$ via random matrix theory following May [3].

A Proof of the Three-Species Inverse Formula

We verify (13) by direct computation. Let $\mathbf{A} = (a_{ij})$ with $a_{ii} = 1$. Expanding $\det(\mathbf{A})$ by the first row:

$$D = 1 \cdot (1 - \alpha_{23}\alpha_{32}) - \alpha_{12}(\alpha_{21} - \alpha_{23}\alpha_{31}) + \alpha_{13}(\alpha_{21}\alpha_{32} - \alpha_{31}),$$

which simplifies to (15). Each entry of $\mathbf{A}^{-1}$ is the transposed cofactor divided by D; all six off-diagonal cofactors are listed in (13). The formula $\mathbf{A} \cdot \mathbf{A}^{-1} = I$ is verified row-by-row using (15). $\square$

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Glossary

CLV Competitive Lotka–Volterra system (1).

Competition matrix A
Matrix of intra- and inter-specific competition coefficients; $A_{ii} = 1$, $A_{ij} = \alpha_{ij} \geq 0$ for $i \neq j$.

Community matrix
Jacobian J of (1) evaluated at an equilibrium; governs local stability via its eigenvalues.

Carrying capacity K_i
Maximum population of species i in the absence of interspecific competition.

Competitive exclusion
Gause's principle: two species competing for the same resource cannot stably coexist unless niche differentiation occurs.

Determinant D $\det(\mathbf{A})$; vanishes iff the interior equilibrium is non-unique or non-existent.

Interior equilibrium $\mathbf{x}^*$
Equilibrium with all $x_i^* > 0$; given by $\mathbf{A}^{-1}\mathbf{K}$ when $D \neq 0$.

Intrinsic growth rate r_i
Per-capita growth rate of species i at low population density.

Limit cycle Closed trajectory in phase space; observed in cyclic RPS competition with unstable interior equilibrium.

MLE Maximum likelihood estimator; for CLV reduces to ordinary least squares on linearised observations.

Neural ODE ODE whose vector field is parameterised by a neural network and trained via the adjoint method [9].

Nullcline Curve in phase space where $\dot{x}_i = 0$; intersection of all N nullclines gives equilibria.

Positive definite
A matrix M is positive definite if $\mathbf{v}^\top M\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$; for symmetric $\mathbf{A}$ equivalent to all eigenvalues positive.

RK4 Fourth-order Runge–Kutta method; global error $\mathcal{O}(h^4)$ (Algorithm 8.1).

Rock-paper-scissors (RPS)
Non-transitive dominance cycle among three species; associated with cyclic α structure.

Saddle equilibrium
Equilibrium with both stable and unstable manifolds; outcome depends sensitively on initial conditions.

Sylvester's criterion
Positive definiteness of a real symmetric matrix is equivalent to all leading principal minors being positive.

The End